Brain Tumor MRI Images Classification

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***Abstract*—Deep learning is a field of AI that gain a lot of interest over the past few years. It was widely applied to several applications and proven to be a powerful AI tool for many of the complex problems. Despite advancements in medical imaging, the gold standard for brain tumor identification remains the histopathological examination of tissue biopsies. However, this traditional approach is inherently invasive, labor-intensive, and susceptible to human oversight. To address these limitations, there is a critical need for automated diagnostic systems. This study explores the implementation of Convolutional Neural Networks (CNNs) to facilitate the multi-class categorization of brain tumors, aiming to provide a faster and more reliable tool for early clinical intervention. In this paper, we used 3 CNNs models for classifying a large clinical dataset into 4 classes Healthy, Pituitary, Meningioma and Glioma. The proposed CNN models are compared with other popular state-of-the-art CNN models such as EfficientNetB0, EfficientNetV2-M, Xception. The main constribution of this project is to test if remove some layers in the models still give high accuracy.**

I. Introduction

Brain tumors represent a serious and potentially life-threatening condition that requires early and accurate diagnosis to improve patient outcomes. Magnetic Resonance Imaging (MRI) is the primary imaging modality used for brain tumor detection due to its high resolution and ability to capture detailed soft tissue structures. However, the manual interpretation of MRI scans is a complex and time-consuming process that depends heavily on the expertise of radiologists, and may lead to variability and diagnostic errors [1].

In recent years, deep learning techniques have significantly advanced the field of medical image analysis, particularly in brain tumor detection and classification. Convolutional Neural Networks (CNNs) have demonstrated strong performance in automatically extracting relevant features from MRI images and identifying tumor patterns with high accuracy [2], [3]. These models have become central to the development of Computer-Aided Diagnosis (CAD) systems, aiming to assist clinicians in improving both efficiency and reliability.

Several studies have explored different CNN-based architectures for brain tumor classification, achieving promising results by leveraging image enhancement techniques, customized network designs, and advanced feature extraction strategies [4], [5]. In addition, large benchmark datasets such as the BRATS dataset have played a crucial role in standardizing evaluation protocols and enabling robust comparisons between different approaches [6].

Despite these advancements, significant challenges remain, including class imbalance, variability in tumor size and appearance, and differences in imaging conditions across datasets. Recent reviews highlight that while deep learning models achieve high performance, their generalization and robustness in real clinical settings remain key limitations [7].

In this study, we propose a deep learning-based framework for brain tumor detection and classification from MRI images. Our approach aims to leverage the strengths of convolutional architectures to improve both classification accuracy and localization capabilities. The objective is to develop an efficient and reliable system while analyzing the impact of different modeling strategies on performance.

Overall, this work contributes to the ongoing research in medical imaging by exploring practical deep learning solutions for brain tumor analysis and identifying potential directions for future improvements in clinical decision-support systems.

II. Dataset

A. Brain tumor and the importance of its classification

A brain tumor is an abnormal growth of cells in the brain. Tumors can be non-cancerous or malignant cancerous, and they may affect brain function by putting pressure on surrounding tissues. Common types of brain tumors include glioma, meningioma, and pituitary tumors, each developing in different parts of the brain and requiring different treatment approaches.

Brain tumor classification is important because it helps doctors identify the type of tumor and choose the most appropriate treatment. Accurate classification can support early diagnosis, improve treatment planning, and increase the chances of successful patient outcomes.

B. About the data

The dataset used in this project is the Brain Tumor MRI Images Classification Dataset from Kaggle.The images are provided in their original form and have not been preprocessed.

The dataset is divided into four classes: glioma tumor, meningioma tumor, pituitary tumor, and no tumor. The images

are organized into training and testing folders, allowing machine learning models to learn from the training data and evaluate performance on the testing data.

C. Data Visualization

The following figure and table illustrate the dataset before preprocessing. Fig. 1 shows representative samples from each class, while Table I presents the corresponding label encoding.

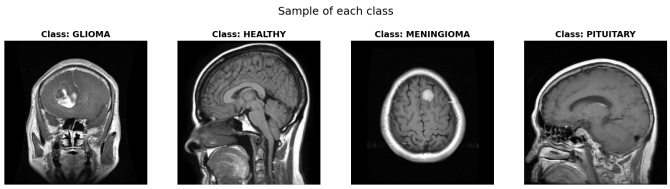

Fig. 1. Sample images from each tumor class (glioma, healthy, meningioma, and pituitary).

TABLE I
SAMPLE DATASET WITH TUMOR CLASS LABELS

Filename	Glioma	Healthy	Meningioma	Pituitary
healthy_0996.jpg	0	1	0	0
pituitary_0822.jpg	0	0	0	1
meningioma_1191.jpg	0	0	1	0
healthy_3874.jpg	0	1	0	0
glioma_3708.jpg	1	0	0	0

D. Data preprocessing

The data preprocessing stage is a critical component of the pipeline, designed to standardize the input and enhance the model's ability to generalize to unseen medical images. Given the inherent variability in MRI scans—such as differences in lighting, contrast, and patient positioning—a multi-step transformation process was implemented to ensure the model focuses on the actual pathological features of the tumors.

As illustrated in the preprocessing analysis, every raw MRI image underwent a resolution standardization to 225×225 pixels. This ensures a consistent input tensor for the EfficientNet V2 and Xception architectures.Following resizing, pixel values were normalized to a range of $[-1, 1]$. This normalization is essential for maintaining numerical stability during the weight update process, preventing large gradient swings and allowing the model to converge more efficiently during the training phase.

To further bolster the model's robustness, a data augmentation strategy was employed, which effectively tripled the volume of the training data. This process is visualized in the "Raw picture" vs. "Increased Contrast" examples. By randomly applying contrast enhancements and horizontal flips, the model is forced to learn invariant features. For instance, the horizontal flip ensures the model does not become biased toward a specific side of the brain, while contrast adjustments simulate variations in MRI scanner settings. This comprehensive preprocessing ensures that the final model is not merely memorizing the training set but is instead learning to identify the core visual signatures of brain tumors.

III. EXPERIMENTAL

All experimental was doing on Kaggle notebook with GPU T100 and Google Colab GPU T4.

IV. EVALUATION - RESULT EXPLANATION

show the metrics, accuracy, result

V. CONCLUSION

This study successfully developed an automated system for brain tumor classification using MRI images through an end-to-end Deep Learning approach. By evaluating several state-of-the-art architectures on the Kaggle Brain Tumor MRI Dataset, a clear distinction in operational efficiency was identified. While the Xception model achieved high accuracy (99.9The experimental results indicate that advanced convolutional neural networks can reach near-perfect precision and recall without the need for manual feature engineering. The high performance across all metrics suggests that the model has successfully learned the complex spatial hierarchies required to distinguish between Glioma, Meningioma, and Pituitary tumors. This robust framework provides a reliable foundation for real-time diagnostic support in clinical settings. Future research will focus on expanding the dataset diversity to further validate the model's reliability across different MRI hardware and protocols.

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